

# COMPASS → SCOUT: Model coherence — Alex score vs decomposable L model

**Date:** 2026-06-11 16:26 (finalized)  
**Created:** 2026-06-11 09:11  
**From:** COMPASS  
**To:** SCOUT  
**Priority:** **Central** — VECTOR cannot draft §3 (minimal model) without your nesting story under the locked plan.

Charles: please route to SCOUT. Answers update near-term plan and VECTOR prose guidance.

## How to respond (naming & location)

| Field    | Rule                                                                                                                                                                                                       |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Location | 3-Master_Plan/                                                                                                                                                                                             |
| Filename | YYYYMMDD_HHMM_SCOUT_to_COMPASS_model_coherence.md — mirror this file ( COMPASS_to_SCOUT_model_coherence → SCOUT_to_COMPASS_model_coherence ); use <b>your local response date/time</b> (24h) as the prefix |
| Example  | 20260611_1700_SCOUT_to_COMPASS_model_coherence.md                                                                                                                                                          |
| Format   | Markdown; address deliverables D1–D11 and §B–§C questions; prose welcome                                                                                                                                   |

**Artifacts / lab scope (separate queue):** reply to 20260611\_1626\_COMPASS\_to\_SCOUT\_questions.md as YYYYMMDD\_HHMM\_SCOUT\_to\_COMPASS.md .

Save each response file in 3-Master\_Plan/ and notify Charles/COMPASS when complete. Do **not** edit this question file unless Charles asks.

## 0. Decision locked (Charles, 2026-06-11)

Charles chose **Path A — manuscript-first** for Summer–Fall 2026 generative priority.

In the reconciliation table below (§D), that is **Path II — Separate layers**:

| Layer                                 | Role in v1 manuscript                                                             | Blocking?                                                            |
|---------------------------------------|-----------------------------------------------------------------------------------|----------------------------------------------------------------------|
| Empirical stylized fact               | Inverted-U on LOO pool quality ( poolq_loo ) — 530 / 538                          | No — already strong                                                  |
| Minimal generative (Alex score)       | Constraint-leg POC; inverted-U readout on pool mean (non-LOO)                     | No — document honestly with axis table                               |
| Empirical decomposition / predictions | Mechanism columns, heterogeneity — same $L = B - D$ story, not a second mechanism | No — needed for §4 predictions                                       |
| Generative LOO-pool-quality match     | Bin-for-bin U on poolq_loo in CELL 10                                             | <b>No</b> — parallel science only; Alex or Charles may elevate later |

**Implication for this document:** COMPASS is **not** asking you to pick a path. Charles locked **Path II**. Your job is to make Path II **scientifically coherent** — explicit nesting, honest axis language, and concrete deliverables VECTOR can cite — not to relitigate whether generative must match LOO pool quality before draft.

**Path I (full nest under B–D with generative LOO U)** remains valuable **parallel** work if time allows; it does **not** gate the first manuscript draft.

## 1. Why this document still exists

Charles raised a **coherence concern** COMPASS shares: the program can read as **two models** (“Alex for curves, multivariate for predictions”) unless someone writes the **one-ontology nesting** clearly.

**Charles’s structural expectation:** one minimal object whose sub-components can be switched on/off:

$$L_{\text{net}} = B(\cdot) - D(\cdot)$$

**Alex’s contribution:** selection score with LOO congestion in the score; inverted-U when conditioned on **pool mean**:

$$S_i = A_i - \lambda \cdot L_{C,\text{LOO}}$$

**Tension (unchanged):** empirical U lives on **LOO pool quality**; current generative knobs mostly show U on **pool mean** and decreasing on LOO pool quality.

Under **locked Path II**, the manuscript must **never** claim the generative sim “explains” the LOO-pool-quality U. It must **still** explain how Alex score + decomposition + empirical U are **one**  $L = B - D$  program — not pragmatics alone.

2. Terminology lock (Charles + COMPASS)

| Term                                | Rule                                                                                                                |
|-------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| Any quantity excluding self         | Name must contain <code>LOO</code> (e.g. LOO pool quality = <code>poolq_loo</code> , LOO congestion = $L_{C,LOO}$ ) |
| Whole-roster average including self | Pool mean / <code>team_mean</code> — not LOO                                                                        |

Empirical stylized fact (530): inverted-U on LOO pool quality ( `poolq_loo` ).  
Generative (538D, current knobs): inverted-U on pool mean; mostly decreasing on LOO pool quality axis.

3. Questions for SCOUT (revised for locked Path II)

Answer in order. Where a question assumed “pick a path,” answer given Path II is locked.

A. Two-track advice under Path II

1. Is “Alex score for generative curve POC; decomposed empirical for predictions” still your recommendation when Path II is the manuscript plan?  
Yes / No / Yes with qualifications — one paragraph. If you disagree with Charles’s lock, say so; COMPASS will note dissent, not override Charles.
2. Map the locked manuscript to the Wang ladder (no optional paths):

| Rung | Label                       | Model object             | What it proves in v1 |
|------|-----------------------------|--------------------------|----------------------|
| 1    | Empirical stylized fact     | ?                        | ?                    |
| 2    | Minimal generative          | Alex score + assignment? | ?                    |
| 3    | Predictions / decomposition | Multivariate empirical?  | ?                    |

3. Under Path II, is Rung 2 allowed to differ from Rung 1 on the conditioning axis (pool mean vs LOO pool quality)? State the exact sentence VECTOR may use in §3 so reviewers do not read “two mechanisms.”

B. Nesting — one ontology (required for Path II prose)

4. Where does the Alex score live in  $L_{net} = B - D$ ?
- a. Operationalization of  $D$  only (constraint / LOO congestion in selection)?

b. Reduced-form substitute for entire  $L_{net}$ ?

c. Separate bundle — if you choose this, Path II fails Charles’s coherence test; explain or pick (a)/(b)

- d. Other — specify.

5. Provide **one** diagram or bullet chain VECTOR can paste into §3:

```
Structural L_net = B - D
→ reduced-form proxy (LOO pool quality)    [Rung 1 axis]
→ Alex score (which term?)                  [Rung 2]
→ empirical mechanism columns (B vs D map)   [Rung 3]
→ generative CELL 10 (knob → term map)
```

6. For each empirical mechanism column you want in predictions (e.g. `congestion_quality`, `peer_perf_sd_loo`, minutes share): label **B**, **D**, or **diagnostic of B–D** — not “extra regressors.”

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### C. Stochastic / bundle differences (539 vs 538D vs decomposition plan)

7. List **every** stochastic layer in: 539 bundled DGP; 538D CELL 10 top- $K$ ; Charles's decomposition plan. Which differences are **scientific** vs **engineering**?

8. Under Path II, what is the **minimal generative claim** VECTOR may make?

- Score equation only?
  - Score + named B/D in prose without generative B term?
  - Something else — one sentence for the claim box.
- 

### D. Path II deliverables (manuscript-critical)

Charles does **not** need generative LOO-pool-quality replication before draft. He **does** need the following from SCOUT:

| ID | Deliverable                                                                                           | COMPASS assumption — confirm or correct        |
|----|-------------------------------------------------------------------------------------------------------|------------------------------------------------|
| D1 | <b>Two-row axis table</b><br>(pool mean → inverted-U; LOO pool quality → current generative readout)  | Required main text or supplement               |
| D2 | <b>Frozen score equation +</b><br>ability-only null<br>reference run                                  | Required                                       |
| D3 | <b>Manuscript Figure 2</b><br>path (empirical LOO U) +<br>generative figure(s) with<br>honest caption | Required                                       |
| D4 | <b>CELL 7+ robustness</b><br>(FE, clustering)                                                         | Deferrable post-draft unless you say hard gate |
| D5 | <b>Generative LOO-pool-<br/>quality match</b>                                                         | <b>Deferred</b> — parallel only                |
| D6 | <b>CELL 4D heterogeneity</b><br>unparked for one<br>prediction                                        | Beneficial — confirm priority                  |

9. For each D1–D6: **confirm**, **reject**, or **revise** the assumption; add **file paths** and **dates** where known.
10. What is the **single next SCOUT coding task** under locked Path II (not “if Path I vs II”)? Be specific: notebook cell, export, or prose artifact.
11. What should SCOUT **freeze** (no more code churn) after that task?

E. Predictions — same story, not a second model

12. State **one** testable prediction sentence implied by **the same**  $L = B - D$  story using decomposed empirical inputs (not ad hoc regression).
- Example shape (you improve):
- “Elite-tier dip in draft rate should be steepest for near-threshold players where LOO congestion is high but LOO pool quality is also high.”*
13. Can that prediction be tested **without** generative LOO-pool-quality U replication? If yes: which cells, which exports, timeline?
14. If you still recommend eventual **Path I** nesting work, name **one** milestone that would most improve Path II prose (e.g. “show  $L_{C,LOO}$  maps to  $D$  in sim”) — **optional**, time-boxed, non-blocking.

4. Relationship to 20260611\_1626\_COMPASS\_to\_SCOUT\_questions.md

| 1626 SCOUT questions (artifacts)     | Status after Charles Path A                                                                                                 |
|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| A1 — $L_Q$ generative gap required?  | Answered: No for v1 (honest axis table + limitations)                                                                       |
| A2 — Both Plot B axes in manuscript? | Still need SCOUT — likely supplement or methods; confirm                                                                    |
| A3 — CELL 7 hard gate?               | Still need SCOUT                                                                                                            |
| B4 — Priority order                  | Prefilled: (e) manuscript support → (c) CELL 7 → (b) 4D → (d) sweep → (a) LOO generative match deferred — confirm or revise |
| B5 — Minimum 4-week deliverable      | Replaced by D10 in this doc                                                                                                 |
| C, D, E in 1626 SCOUT questions      | Still open — answer in 20260611_1626_COMPASS_to_SCOUT_questions.md or here                                                  |

This document is authoritative for model coherence and nesting; 20260611\_1626\_COMPASS\_to\_SCOUT\_questions.md remains authoritative for artifacts, PERF\_METRIC, 535/539 status.

5. What COMPASS will do with answers

- Q3 locked in PROJECT\_STATUS\_AND\_NEAR\_TERM\_PLAN.md (Charles Path A).
- Brief VECTOR on allowed §3 mechanism prose from your nesting chain (§B.5) and minimal claim (§C.8).
- Set Phase B SCOUT tasks from D1–D11, not from generative LOO match.

6. Charles’s position (engage; do not argue past)

Charles believes:

- A minimal model should decompose into sub-components, not run two parallel mechanisms for “curves” vs “predictions.”
- He chose Path II because Summer–Fall 2026 requires draftable honesty, not generative perfection on every axis.
- He still needs explicit nesting — Path II is not permission for “two models” rhetoric.

Please respond with enough detail that VECTOR can draft §3 without contradictions.

End model coherence questions (Path II locked). Save as YYYYMMDD\_HHMM\_SCOUT\_to\_COMPASS\_model\_coherence.md in 3-Master\_Plan/ (see § How to respond).